

Exact Certification of the 11-Dimensional Kissing-Number Upper Bound

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Abstract

We describe a proof-oriented implementation of the degree-17 Bachoc–Vallentin three-point semidefinite program for the 11-dimensional kissing number. The reported numerical objective is 868.82650. The certification target fixes the objective at 86899/100. A compact exact rational certificate at that objective passes primary Julia verification and independent release-mode replay, establishing $\tau_{11} \leq 868$. The package verifies all 70 matrix blocks and every exact coefficient identity within the declared residual spaces. The final prior-art refresh located no equivalent public certificate. The release procedure independently replays the certificate before publishing immutable tag-bound assets.

1 Problem

The kissing number τ_n is the largest number of nonoverlapping unit spheres in $\mathbb{R}^n$ that can touch one central unit sphere. Equivalently, it is the largest spherical code in S^{n-1} whose distinct vectors have inner product at most $1/2$.

The Bachoc–Vallentin three-point method gives semidefinite-programming upper bounds for spherical codes. A degree-17 computation in dimension 11 reports the numerical value 868.82650, but the source paper states that its table was not rigorously verified.

2 Certification Target

The project replaces objective minimization by the exact equality

$$\text{objective} = \frac{86899}{100} = 868.99.$$

An exact feasible solution at this value implies

$$\tau_{11} \leq 868,$$

because τ_{11} is an integer.

3 Reduced Three-Point Bound

Let P_k^n denote the normalized Gegenbauer polynomial with $P_k^n(1) = 1$, and define

$$\Delta_s = \{(u, v, t) \in [-1, s]^3 : 1 + 2uv t - u^2 - v^2 - t^2 \geq 0\}.$$

For nonnegative $a_0, \dots, a_r$ and positive-semidefinite $F_0, \dots, F_d$, put

$$G(u) = \sum_{k=0}^r a_k P_k^n(u), \quad H(u, v, t) = \sum_{k=0}^d \langle \bar{Y}_k^n(u, v, t), F_k \rangle.$$

The reduced three-point bound states that

$$G(u) + 3H(u, u, 1) \leq -1 \quad (u \in [-1, s])$$

and

$$H(u, v, t) \leq 0 \quad ((u, v, t) \in \Delta_s)$$

imply

$$A(n, \arccos s) \leq M := 1 + \sum_{k=0}^r a_k + \langle J, F_0 \rangle.$$

For completeness, let C be a spherical code with $N = |C|$. Positivity of the three-point kernels gives

$$T = \sum_{x, y, z \in C} H(x \cdot z, y \cdot z, x \cdot y) \geq 0.$$

Separating triples by their number of distinct points and applying the two displayed inequalities gives

$$T \leq NH(1, 1, 1) - N(N-1) - \sum_{k=0}^r a_k \sum_{x \neq y} P_k^n(x \cdot y).$$

Schoenberg positivity implies

$$\sum_{x \neq y} P_k^n(x \cdot y) \geq -N.$$

Since $H(1, 1, 1) = \langle J, F_0 \rangle$, it follows that $0 \leq T \leq N(M - N)$ and hence $N \leq M$.

This is the reduced formulation of Dostert, de Laat, and Moustrou [4] and the mixed-degree formulation used by Leijenhorst and de Laat. It is not described here as a literal transcription of Bachoc–Vallentin Theorem 4.2, which also contains an auxiliary 2×2 block.

4 Proof Architecture

The certificate chain has four stages:

1. reproduce the reported high-precision numerical bound;
2. solve the fixed-objective feasibility problem with positive slack;
3. project the numerical solution into the exact rational affine space;
4. verify every affine identity and rational positive-semidefinite block with two independent implementations.

The primary implementation uses Julia and Nemo together with the pinned solver package `ClusteredLowRankSolver.jl`. The release certificate uses a deterministic interchange format rather than Julia serialization.

The trivariate residuals lie in the symmetric polynomial space generated by

$$(u + v + t)^a(uv + ut + vt)^j(uxt)^k, \quad a + 2j + 3k \leq 34.$$

This space has dimension 1461. The canonical 1461-point evaluation matrix has rank 1461 modulo 65521. Therefore it has full rank over the rationals, and exact zero residuals at all canonical samples imply coefficient equality within this declared space.

5 Certificate Verification

The rational problem generator and fixed-objective mode pass unit tests. A degree-2 pilot in dimension 3 reproduces objective 14. A fixed-objective pilot rounds to an exact rational feasible solution with zero affine residuals and 16 positive-semidefinite blocks verified in exact arithmetic. The exact verifier also rejects nonrational scalar types, wrong block layouts, and objective mismatches, and it reconstructs the canonical problem rather than trusting a serialized problem object.

A size audit rejects direct publication of the sampled affine system. The 1,663,458 nonzero three-point kernel terms alone require at least 93,153,648 bytes in that representation, and the complete expansion has an upper bound of 505,535,295 terms. The replacement certificate stores each exact rational symmetric upper triangle. The independent verifier proves strict positivity of release matrices by directed fixed-point interval Cholesky. A solver-produced degree-2 pilot reconstructs exactly in Julia and passes complete independent Python coefficient replay.

The independent verifier uses only standard-library rational arithmetic. Its 56 tests cover the complete 70-block target layout, strict binary parsing, exact objective and polynomial identities, malformed inputs, package-manifest bindings, provenance recomputation, and resource limits. Non-degenerate cross-language checks compare Julia and Python on all 35 scalar kernels, three contractions for each of the 18 degree-17 F blocks, both interval SOS blocks, and three contractions for each of the 15 three-point SOS families at degree 6.

The retained 256-bit dimension-11 run returned status `pd0pt` and objective

$$868.826499507902298742415116214534666 \dots$$

The accepted 384-bit fixed-objective recovery has objective error about 1.17×10^{-68} , maximum sampled affine residual about 1.63×10^{-71} , and minimum matrix eigenvalue about 9.58×10^{-15} . Portable exact affine projection and primary exact verification now pass all 1,497 identities and all 70 strict-positive matrix blocks. The resulting KNWIT003 witness is 14,850,075 bytes and its manifest is 12,458 bytes. Export-time captured-byte replay and a second installed-file release-mode replay verify all 70 blocks, all 35 univariate coefficients, and all 1,461 symmetric trivariate coefficients. Deterministic compression produces a 14,565,324-byte package. The standalone replay took 907.16 seconds and 187,596,800 bytes maximum RSS, below the two-hour and 16 GiB limits. Together with the theorem bridge, these checks establish $\tau_{11} \leq 868$.

6 Release Procedure

The mathematical, independent-verification, compactness, and resource gates pass. The manuscript also builds reproducibly and passes visual inspection. The release workflow performs a clean hosted

replay, rebuilds the paper from the tagged source, verifies every release-asset digest, and only then publishes the immutable GitHub release. Zenodo archival follows that public release and uses the same four byte-verified files.

References

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